Organic Chemistry Cumulative Examination
December 5, 1992

Problem 1 is worth a maximum of 40 points; problem 2, 60 points. A passing grade is about 55 points.

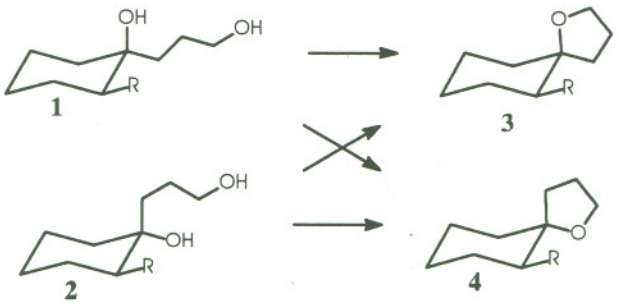
1. (Gleiter, R.; Merger, R.; Nuber, B. *J. Am. Chem. Soc.* **1992**, *114*, 8921.) Determine the structure of compounds **A** and **B** from the following data.

   A: Molecular formula: $C_{12}H_{12}$
   IR: 2276, 2210, 1640 cm^{-1}
   ^1H NMR: δ 4.9 (s, 4H), 3.0 (s, 8H)
   ^{13}C NMR: δ 138.9, 115.5, 82.1, 27.8

   B: Molecular formula: $C_{12}H_{12}$
   IR: 2274, 2210, 1643 cm^{-1}
   ^1H NMR: δ 5.68 (t, 4H), 2.87 (d, 8H)
   ^{13}C NMR: δ 127.2, 78.4, 17.2

   Attached are some spectral correlation charts which may be helpful.

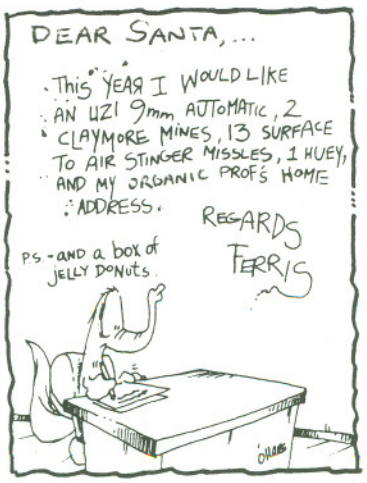
2. (Negri, J. T.; Paquette, L. A. *J. Am. Chem. Soc.* **1992**, *114*, 8835.) This paper is about the acid-catalyzed cyclodehydration of 2-substituted 1-(3-hydroxypropyl)cyclohexanols to form spirocyclic tetrahydrofurans.



Several R groups were used, namely OCH_3, C_6H_5, and CH3. Isotopic labeling experiments were performed where the starting diols incorporate about 40% ^{18}O at the C-1 of the cyclohexane. The following results were observed.

starting diol	% ^{18}O of starting diol	product ratio, **3** : **4**	% ^{18}O in **3**	% ^{18}O in **4**
1-OCH_3	40	96 : 4	40	11
2-OCH_3	39	8 : 92	16	39
1-C_6H_5	45	67 : 33	23	0
2-C_6H_5	45	55 : 45	0	21
1-CH_3	39	40 : 60	10	0
2-CH_3	38	55 : 45	0	8

Analyze these results in terms of the mechanism(s) of the reaction, with particular attention to the electronic and steric effects of the R groups.



Bond Type	Absorption Region (cm^{-1})
C—C, C—O, C—N	1300–800
C=C, C=O, C=N, N=O	1900–1500
C≡C, C≡N	2300–2000
C—H, O—H, N—H	3800–2700

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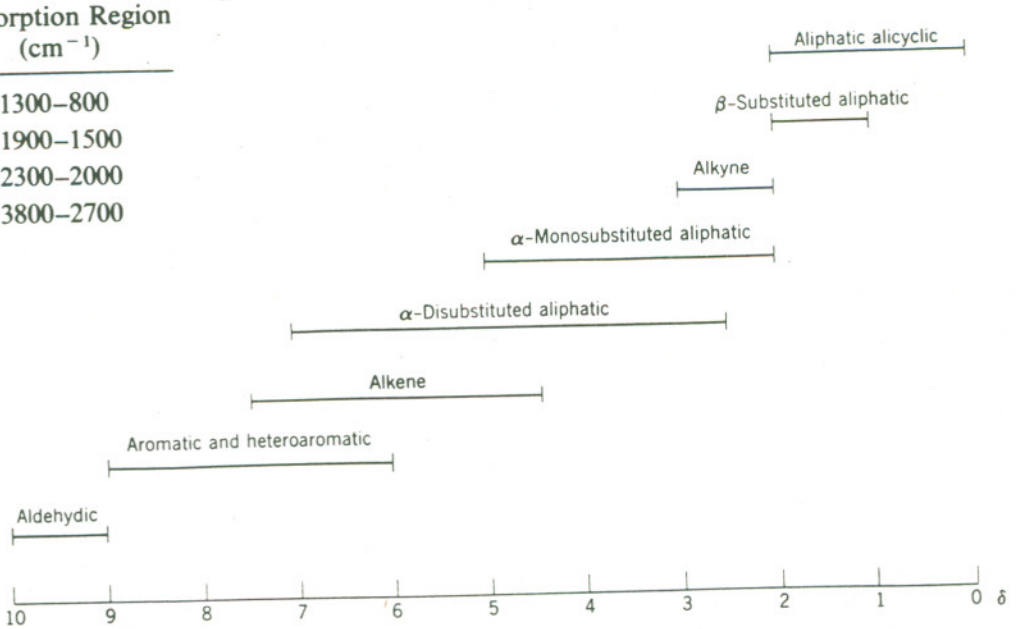

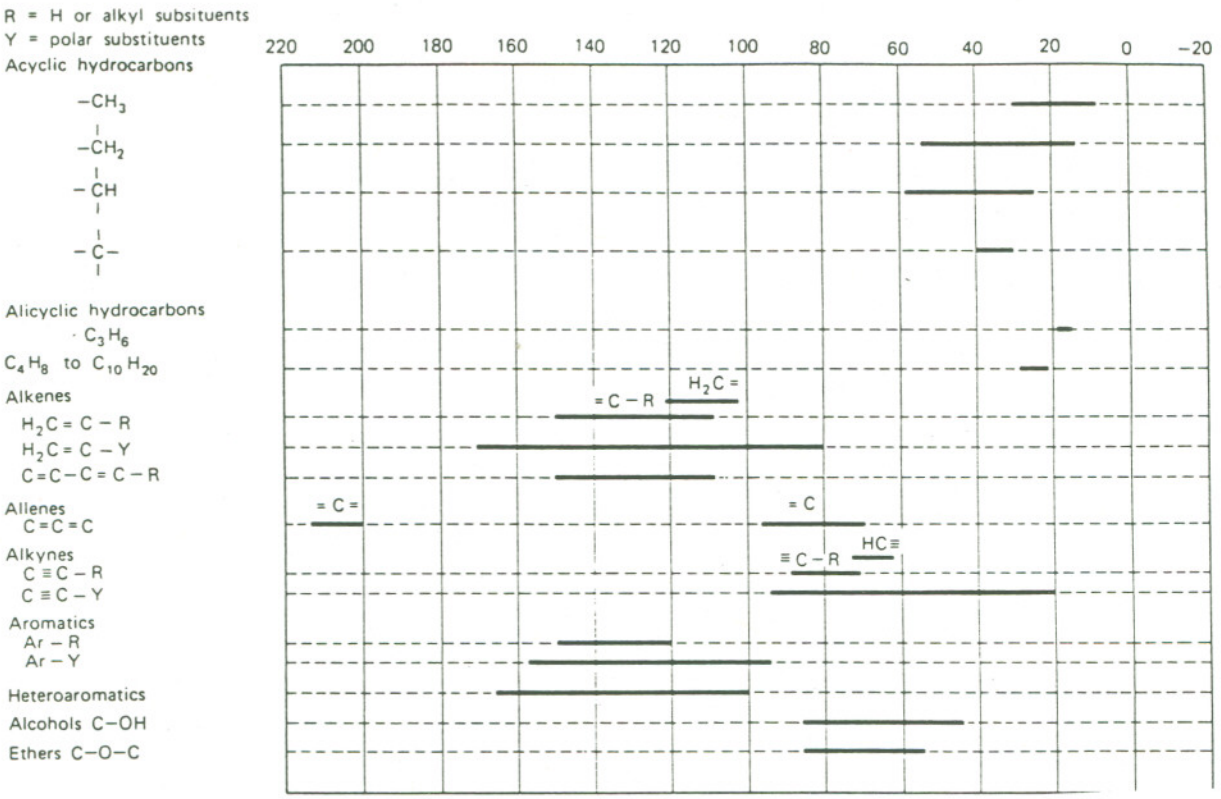

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TABLE 1.3 Field effects of various groups relative to hydrogen

The groups are listed approximately in order of decreasing strength for both −I and +I groups

(e⁻ donating) +I		−I (e⁻ withdrawing)	
O^-	NR_3^+	COOH	OR
COO^-	SR_2^+	F	COR
CR_3	NH_3^+	Cl	SH
CHR_2	NO_2	Br	SR
CH_2R	SO_2R	I	OH
CH_3	CN	OAr	C≡Cr
D	SO_2Ar	COOR	Ar
			$CH{=}CR_2$